Swami Shreeji !

VIT

Vellore Institute of Technology

Vellore – 632014, Tamil Nadu, India.

SCHOOL OF ADVANCED SCIENCES
DEPARTMENT OF MATHEMATICS
WINTER SEMESTER – 2023~2024

Continuous Assessment Test – I

Programme Name & Branch : B.Tech. (All Branches) | Duration : 90 Minutes
Course Code : BMAT205L | Maximum Marks : 50
Course Name : Discrete Mathematics and Graph Theory | Slot : C1+TC1+TCC1

General Instruction(s): —

Answer ALL the Questions

(5 × 10 = 50 Marks)

1. Show that the following statements are inconsistent.
If Jack gets his UG degree, then he will go for a job.
If Jack goes for a job, then he will get married soon.
If Jack goes for higher study, then he will not get married.
Jack gets his UG degree and goes for higher study. **[10 M]**

2. Obtain the principal disjunctive and conjunctive normal forms of the following statement:

$$\Big(P \to (Q \wedge R)\Big) \vee \Big(\neg P \to (\neg Q \wedge \neg R)\Big).$$

[10 M]

3. Show that the conclusion $(\forall x)\Big(F(x) \longrightarrow \neg S(x)\Big)$ follows logically from the premises $(\exists x)\Big(F(x) \wedge S(x)\Big) \longrightarrow (\forall y)\Big(M(y) \to W(y)\Big)$ and $(\exists y)\Big(M(y) \wedge \neg W(y)\Big)$.

[10 M]

4. (i) For any commutative monoid $\langle M, * \rangle$, show that the set of idempotent elements of M forms a submonoid.

(ii) Give an example of a semigroup but not a monoid. Also give an example of a monoid but not a group. Justify the answers.

[10 M]

5. Show that $G = \left\{ \begin{pmatrix} a & a \\ a & a \end{pmatrix} : a \neq 0 \text{ and } a \in \mathbb{R} \right\}$ is a group under usual matrix multiplication. **[10 M]**

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